

HORNSBY GIRLS' HIGH SCHOOL



YEAR 11 MATHEMATICS 2010 Preliminary Assessment Task 1

Student Number: _____

Time Allowed: 65 minutes

Instructions:

- Attempt all questions
- Start a new page for each question
- The marks for each question are indicated
- Show all necessary working
- Use blue or black pen
- Marks may be deducted for untidy or badly arranged work
- Board approved calculators may be used

Outcomes Assessed:

- P1 Demonstrates confidence in obtaining realistic solutions to problems
- P2 Provides reasoning to support conclusions in the correct context
- P3 Performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4 Chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
- P5 Understands the concept of a function and the relationship between a function and its graph

Marking Scheme:

Outcomes	Questions				Totals
	Q1	Q2	Q3	Q4	
Basic Algebra and Arithmetic	/8	/10	/9	/10	/37
Functions and Graphs	Q5	/8	Q6.	/8	/16
Total					/53

This assessment task constitutes 20% of the final Preliminary Course Assessment.

Question 1 (8 Marks) START A NEW PAGE**Marks**

- a) Evaluate correct to 2 decimal places: $\frac{2.3+19^2}{\sqrt{4.2 \times 0.16}}$ 2
- b) Express $0.\dot{3}\dot{3}\dot{7}$ as a common fraction 2
- c) Rationalise the denominator of $\frac{2}{\sqrt{10}-\sqrt{5}}$ 2
- d) Given $x = \sqrt{3}$, $y = 8\sqrt{3}$ find in its simplest form, the exact value of: $\sqrt[3]{\frac{x}{y}}$ 2

Question 2 (10 Marks) START A NEW PAGE**Marks**

- a) Expand and simplify $(\sqrt{x+y}-\sqrt{x})(\sqrt{x+y}+\sqrt{x})$ 2
- b) Expand then simplify $(rs+2t)^2$ 2
- c) If $F = W - kV$, find V (correct to 1 decimal place) if $F = 98$, $W = 146$ and $k = 5.2$ 2
- d) Factorise $(2a-2)x+2(1-a)y$ 2
- e) Factorise $3x^2-11x+6$ 2

Question 3 (9 Marks) START A NEW PAGE**Marks**

- a) Solve $x^2+6x-2=0$ 2
- b) Solve $1-x+2x=-2x-x-1$ 2
- c) Solve $4-\frac{2}{6x+8}=6$ 2
- d) Simplify $\frac{x^3-8}{3x^2+6x+12} \div \frac{x^3-2x^2}{3}$ 3

Question 4 (10 Marks)**START A NEW PAGE****Marks**

a) Solve $3x + 5 \leq -1$

2

b) Solve $\frac{2x-1}{3} - \frac{3x-2}{4} < \frac{1}{2}$

3

c) Solve $|2x-3| \leq 5$

2

d) Solve simultaneously:

$$6x + 5y = -32$$

$$3x - 2y = 2$$

3**Question 5 (8 Marks)****START A NEW PAGE****Marks**

a) For the function $f(x) = x + \frac{1}{x}$

i) Find the value of $f\left(\frac{1}{2}\right)$

2

ii) Show that $f(a) \times f(b) - f(ab) = \frac{a}{b} + \frac{b}{a}$

3

b)
$$f(x) = \begin{cases} x+1, & x \geq 1 \\ -1, & -1 < x < 1 \\ 1-x, & x \leq -2 \end{cases}$$

Find the value of $f(2) + f(-2) + f(0)$

3**Question 6 (8 Marks)****START A NEW PAGE****Marks**

a) Sketch $y = x^2 - 6x$ showing the x and y intercepts and the vertex

2

b) Show that the function $y = \frac{1}{x^2-1} - \sqrt{|x|}$ is an even function

2

c) i) Sketch $(x-2)^2 + (y-5)^2 = 16$

2

ii) Find the domain and range of $(x-2)^2 + (y-5)^2 = 16$

2**END OF PAPER**

Answers

Q1

$$a) \frac{2.3 + 19^2}{\sqrt{4.2 \times 0.16}} = 443.18$$

$$b) \text{ Let } x = 0.337373 \dots$$

$$\therefore 100x = 33.7373 \dots$$

$$100x - x = 33.4$$

$$99x = 33.4$$

$$x = \frac{33.4}{99}$$

$$= \frac{334}{990}$$

$$= \frac{167}{495}$$

$$c) \frac{2}{\sqrt{10} - \sqrt{5}} \times \frac{(\sqrt{10} + \sqrt{5})}{(\sqrt{10} + \sqrt{5})} = \frac{2(\sqrt{10} + \sqrt{5})}{10 - 5}$$

$$= \frac{2\sqrt{10} + 2\sqrt{5}}{5}$$

$$d) \sqrt[3]{\frac{x}{y}} = \sqrt[3]{\frac{\sqrt[3]{x}}{8\sqrt[3]{x}}}$$

$$= \sqrt[3]{\frac{1}{8}}$$

$$= \frac{1}{2}$$

$$Q2. a) (\sqrt{x+y} - \sqrt{x})(\sqrt{x+y} + \sqrt{x})$$

$$= (\sqrt{x+y})^2 - (\sqrt{x})^2$$

$$= x + y - x$$

$$= y$$

$$b) (rs + 2t)^2 = r^2s^2 + 4rst + 4t^2$$

$$98 = 146 - 5.2V$$

$$5.2V = 48$$

$$V = 9.2$$

$$d) (2a-2)x + 2(1-a)y$$

$$= 2(a-1)x + 2(1-a)y$$

$$= 2(a-1)x - 2(a-1)y$$

$$= 2(a-1)(x-y)$$

$$e) 3x^2 - 11x + 6 \quad \begin{array}{l} p: 18 \\ s: -11 \\ f: -9, -2 \end{array}$$

$$= 3x^2 - 9x - 2x + 6$$

$$= 3x(x-3) - 2(x-3)$$

$$= (3x-2)(x-3)$$

$$Q3. a) x^2 + 6x - 2 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-6 \pm \sqrt{36 - 4 \cdot 1 \cdot -2}}{2}$$

$$= \frac{-6 \pm \sqrt{44}}{2}$$

$$= \frac{-6 \pm 2\sqrt{11}}{2}$$

$$= -3 \pm \sqrt{11}$$

$$b) 1 - x + 2x = -2x - x - 1$$

$$4x = -2$$

$$x = -\frac{1}{2}$$

$$c) 4 - \frac{2}{6x+8} = 6$$

$$-\frac{2}{6x+8} = 2$$

$$-2 = 2(6x+8)$$

$$-1 = 6x+8$$

$$6x = -9$$

$$x = -\frac{3}{2}$$

$$d) \frac{x^2 - 8}{3x^2 + 6x + 12} \div \frac{x^2 - 2x^2}{3}$$

$$= \frac{(x-2)(x^2 + 2x + 4)}{3(x^2 + 2x + 4)} \times \frac{3}{x^2(x-2)}$$

$$= \frac{1}{x^2}$$

Q4 a) $3x + 5 \leq -1$

$$3x \leq -6$$

$$x \leq -2$$

b) $\frac{2x-1}{3} - \frac{3x-2}{4} < \frac{1}{2}$

$$4(2x-1) - 3(3x-2) < 6$$

$$8x - 4 - 9x + 6 < 6$$

$$-x < 4$$

$$x > -4$$

c) $|2x-3| \leq 5$

$$2x-3 \leq 5 \quad \text{or} \quad -(2x-3) \leq 5$$

$$2x \leq 8 \quad -2x+3 \leq 5$$

$$x \leq 4 \quad -2x \leq 2$$

$$x \geq -1$$

$$\therefore -1 \leq x \leq 4$$

d) $6x + 5y = -32$ ①

$$3x - 2y = 2$$
 ②

② $\times 2$: $6x - 4y = 4$ ③

① - ③: $9y = -36$

$$y = -4$$

Sub $y = -4$ in ②:

$$3x - 2(-4) = 2$$

$$3x + 8 = 2$$

$$3x = -6$$

$$x = -2$$

Q5. a) $f(x) = x + \frac{1}{x}$

i) $f\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{1}{\frac{1}{2}}$

$$= 2\frac{1}{2}$$

ii) $f(a) \times f(b) - f(ab)$

$$= \left(a + \frac{1}{a}\right)\left(b + \frac{1}{b}\right) - \left(ab + \frac{1}{ab}\right)$$

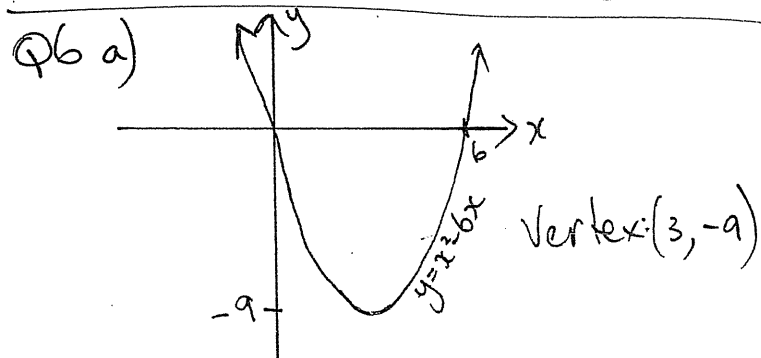
$$= \cancel{ab} + \frac{a}{b} + \frac{b}{a} + \frac{1}{ab} - \cancel{ab} - \frac{1}{ab}$$

$$= \frac{a}{b} + \frac{b}{a}$$

b) $f(2) = 2 + 1 = 3$ $f(-2) = 1 - (-2) = 3$

$$f(0) = -1$$

$$\therefore f(2) + f(-2) + f(0) = 3 + 3 + (-1) = 5$$



b) $f(-x) = \frac{1}{(-x)^2 - 1} - \sqrt{|-x|}$

$$= \frac{1}{x^2 - 1} - \sqrt{x}$$

$$= f(x) \quad \therefore \text{Even.}$$

